

CCPS – Module 3 Cheat Sheet

Parameter Estimation, System Identification and Stochastic Processes

MAP-oriented formula and procedure sheet

Core strategy:

Recognize problem → write correct model → choose formula → calculate → check

1 1. Least Squares – Universal Template

Whenever

$$y(k) = \varphi^\top(k)\theta + e(k),$$

identify

θ = unknown parameter vector, $\varphi(k)$ = known regressor vector.

Stack the measurements:

$$Y = \begin{bmatrix} y(1) \\ \vdots \\ y(N) \end{bmatrix}, \quad \Phi = \begin{bmatrix} \varphi^\top(1) \\ \vdots \\ \varphi^\top(N) \end{bmatrix}.$$

Then

$$Y = \Phi\theta + E.$$

Least-squares cost:

$$J(\theta) = \sum_{k=1}^N [y(k) - \varphi^\top(k)\theta]^2 = \|Y - \Phi\theta\|^2$$

Differentiate:

$$\frac{\partial J}{\partial \theta} = -2\Phi^\top(Y - \Phi\theta).$$

Set equal to zero:

$$\Phi^\top\Phi\hat{\theta} = \Phi^\top Y.$$

Therefore:

$$\hat{\theta}_{LS} = (\Phi^\top\Phi)^{-1}\Phi^\top Y$$

or

$$\hat{\theta}_{LS} = \left[\sum_{k=1}^N \varphi(k)\varphi^\top(k) \right]^{-1} \sum_{k=1}^N \varphi(k)y(k).$$

Always check signs in the regressor. If the model contains $-a y(k-1)$, the regressor contains $-y(k-1)$, not $y(k-1)$.

Uniqueness:

$$\text{rank}(\Phi) = n_\theta \iff \Phi^\top\Phi \text{ is nonsingular.}$$

For a square matrix:

$$\det(\Phi^\top\Phi) \neq 0 \implies \text{unique LS estimate.}$$

Procedure:

1. Write $y(k) = \varphi^\top(k)\theta + e(k)$.
2. Identify $\varphi(k)$ and θ .
3. Calculate $\Phi^\top\Phi$.
4. Calculate $\Phi^\top Y$.
5. Solve the normal equations.
6. Check rank/determinant if identifiability is asked.

2 2. Scalar Least Squares

For

$$y(k) = \theta x(k) + e(k),$$

immediately use

$$\hat{\theta} = \frac{\sum x(k)y(k)}{\sum x^2(k)}$$

Normal equation:

$$\hat{\theta} \sum x^2(k) = \sum x(k)y(k)$$

Residual:

$$\hat{e}(k) = y(k) - \hat{\theta}x(k)$$

Residual sum of squares:

$$J(\hat{\theta}) = \sum \hat{e}^2(k)$$

3 3. Dynamic Models / ARX

For

$$y(k) = -a y(k-1) + b u(k-1) + e(k),$$

write

$$y(k) = \begin{bmatrix} -y(k-1) & u(k-1) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + e(k).$$

Thus:

$$\varphi(k) = \begin{bmatrix} -y(k-1) \\ u(k-1) \end{bmatrix}, \quad \theta = \begin{bmatrix} a \\ b \end{bmatrix}$$

The normal equations are

$$\begin{bmatrix} \sum y^2(k-1) & -\sum y(k-1)u(k-1) \\ -\sum y(k-1)u(k-1) & \sum u^2(k-1) \end{bmatrix} \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} -\sum y(k)y(k-1) \\ \sum y(k)u(k-1) \end{bmatrix}$$

$$\text{model} \rightarrow \varphi, \theta \rightarrow \Phi^\top \Phi \rightarrow \Phi^\top Y \rightarrow \hat{\theta}$$

4 4. Information Matrix and Identifiability

Define

$$R_N = \sum_{k=1}^N \varphi(k)\varphi^\top(k) = \Phi^\top \Phi$$

Unique identifiability:

$$\text{rank}(R_N) = n_\theta$$

For a square matrix:

$$\det(R_N) \neq 0 \implies \text{parameters uniquely identifiable.}$$

If

$$\det(R_N) = 0,$$

then

$$\text{parameters cannot be uniquely identified.}$$

Typical trap:

$$\varphi(k) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \forall k$$

gives

$$R_N = N \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

which is singular.

5 5. Persistent Excitation

Main principle:

$$\text{rich input} \rightarrow \text{rich regressors} \rightarrow R_N \text{ full rank} \rightarrow \text{identifiability.}$$

For

$$\varphi(k) = \begin{bmatrix} u(k) \\ u(k-1) \end{bmatrix},$$

a constant input

$$u(k) = 1$$

gives

$$\varphi(k) = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so the two regressors are linearly dependent.

$$\text{Constant input may fail to identify multiple dynamic parameters.}$$

For experiment design, seek sufficient variation over the relevant frequency range while avoiding actuator saturation, deadzones and excessive noise.

6 6. Recursive Least Squares

Without forgetting:

$$K(k) = \frac{P(k-1)\varphi(k)}{1 + \varphi^\top(k)P(k-1)\varphi(k)}$$

Prediction error:

$$\varepsilon(k) = y(k) - \varphi^\top(k)\hat{\theta}(k-1)$$

Parameter update:

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\varepsilon(k)$$

Covariance update:

$$P(k) = [I - K(k)\varphi^\top(k)]P(k-1)$$

Memorize the order:

$$P, \varphi \rightarrow K \rightarrow \varepsilon \rightarrow \hat{\theta} \rightarrow P_{\text{new}}$$

7 7. RLS with Forgetting Factor

With $0 < \lambda \leq 1$:

$$K(k) = \frac{P(k-1)\varphi(k)}{\lambda + \varphi^\top(k)P(k-1)\varphi(k)}$$

$$\varepsilon(k) = y(k) - \varphi^\top(k)\hat{\theta}(k-1)$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\varepsilon(k)$$

$$P(k) = \frac{1}{\lambda} [I - K(k)\varphi^\top(k)]P(k-1)$$

Effective memory:

$$N_{\text{eff}} \approx \frac{1}{1 - \lambda}$$

Therefore:

$$\lambda \approx 1 - \frac{1}{N_{\text{eff}}}$$

Smaller λ :

$$\text{shorter memory} \rightarrow \text{more recent-data weighting} \rightarrow \text{faster tracking} \rightarrow \text{higher variance.}$$

Example:

$$\lambda = 0.98 \quad \implies \quad N_{\text{eff}} \approx 50.$$

8 8. Matrix Inversion Lemma

Important identity:

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(DA^{-1}B + C^{-1})^{-1}DA^{-1}$$

In RLS, it is used to update an inverse information matrix efficiently after adding a new rank-one data contribution.

9 9. Stochastic Processes

Mean:

$$\mu_e(k) = E[e(k)]$$

Variance:

$$\sigma_e^2(k) = E[(e(k) - \mu_e(k))^2]$$

For zero mean:

$$\sigma_e^2 = E[e^2(k)]$$

Autocovariance:

$$R_e(k_1, k_2) = E[(e(k_1) - \mu_e(k_1))(e(k_2) - \mu_e(k_2))]$$

For zero mean:

$$R_e(k_1, k_2) = E[e(k_1)e(k_2)]$$

Second-order white noise:

$$E[e(k)] = 0, \quad E[e^2(k)] = \sigma_e^2, \quad E[e(k)e(j)] = 0, \quad k \neq j.$$

To prove that a process is not white, it is enough to find one nonzero lagged covariance.

10 10. Colored Noise – Standard Exam Pattern

Suppose

$$v(k) = e(k) + ce(k-1),$$

where $e(k)$ is zero-mean white noise with variance σ_e^2 .

Mean:

$$E[v(k)] = 0.$$

At lag one:

$$v(k-1) = e(k-1) + ce(k-2).$$

Then

$$\begin{aligned} E[v(k)v(k-1)] &= E[(e(k) + ce(k-1))(e(k-1) + ce(k-2))] \\ &= cE[e^2(k-1)] \\ &= c\sigma_e^2. \end{aligned}$$

Therefore:

$$c \neq 0 \implies v(k) \text{ is colored.}$$

11 11. LS Bias with Colored Noise

True model:

$$y(k) = \theta_0^\top \varphi(k) + v(k).$$

LS estimate:

$$\hat{\theta} = \left(\sum \varphi \varphi^\top \right)^{-1} \sum \varphi y.$$

Substituting:

$$\hat{\theta} = \theta_0 + \left(\sum \varphi \varphi^\top \right)^{-1} \sum \varphi v.$$

Therefore the critical condition is

$$E[\varphi(k)v(k)] = 0.$$

If

$$E[\varphi(k)v(k)] = 0,$$

LS can be asymptotically consistent under the relevant assumptions.

If

$$E[\varphi(k)v(k)] \neq 0,$$

LS can be asymptotically biased.

Important dynamic-model mechanism:

$$y(k-1) \text{ contains previous noise}$$

and colored noise contains previous white-noise samples, so

$$y(k-1) \text{ and } v(k) \text{ can become correlated.}$$

12 12. Maximum Likelihood

For independent observations:

$$L(\theta) = \prod_{i=1}^N p(y_i | \theta)$$

Log-likelihood:

$$\ell(\theta) = \log L(\theta)$$

Because logarithm is monotonic:

$$\arg \max_{\theta} L(\theta) = \arg \max_{\theta} \ell(\theta)$$

Procedure:

1. Write the probability density.
2. Form the likelihood.
3. Take log.
4. Differentiate.
5. Set derivative to zero.
6. Solve for the parameter.
7. Check that the solution is a maximum.

13 13. Gaussian Sensor Fusion

For independent measurements

$$y_i = \theta + e_i, \quad e_i \sim \mathcal{N}(0, \sigma_i^2),$$

the ML estimate is

$$\hat{\theta}_{\text{ML}} = \frac{\sum_i y_i / \sigma_i^2}{\sum_i 1 / \sigma_i^2}$$

Each measurement has precision

$$w_i = \frac{1}{\sigma_i^2}.$$

Therefore:

$$\text{smaller variance} \rightarrow \text{larger weight.}$$

For two sensors:

$$\hat{\theta}_{\text{ML}} = \frac{y_1 / \sigma_1^2 + y_2 / \sigma_2^2}{1 / \sigma_1^2 + 1 / \sigma_2^2}$$

For equal variances:

$$\hat{\theta}_{\text{ML}} = \frac{1}{N} \sum_{i=1}^N y_i.$$

14 14. Gaussian Mean and Variance

If

$$y_i = \theta + e_i, \quad e_i \sim \mathcal{N}(0, \sigma^2),$$

independently:

$$\hat{\theta}_{\text{ML}} = \frac{1}{N} \sum_{i=1}^N y_i$$

and

$$\text{Var}(\hat{\theta} - \theta) = \frac{\sigma^2}{N}.$$

For zero-mean Gaussian data with unknown variance:

$$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{N} \sum_{i=1}^N y_i^2$$

For the Gaussian ML variance estimate, the denominator is

$$N$$

not $N - 1$.

If

$$\frac{\sigma^2}{N} < \varepsilon,$$

then

$$N > \frac{\sigma^2}{\varepsilon}.$$

15 15. General Gaussian ML

For

$$Y \sim \mathcal{N}(\mu(\theta), \Sigma),$$

$$L(\theta) \propto |\Sigma|^{-1/2} \exp \left[-\frac{1}{2} (Y - \mu(\theta))^{\top} \Sigma^{-1} (Y - \mu(\theta)) \right].$$

Thus ML minimizes

$$(Y - \mu(\theta))^{\top} \Sigma^{-1} (Y - \mu(\theta)).$$

If observations are correlated, do **not** multiply their marginal PDFs as if they were independent.

16 16. Rayleigh ML

For the Rayleigh density:

$$p(v \mid \sigma) = \frac{v}{\sigma^2} \exp \left(-\frac{v^2}{2\sigma^2} \right), \quad v \geq 0.$$

For independent observations:

$$\hat{\sigma}^2 = \frac{1}{2N} \sum_{i=1}^N v_i^2.$$

17 17. Consistency, Efficiency and Fisher Information

Consistency:

$$\hat{\theta}_N \rightarrow \theta_0 \quad \text{as} \quad N \rightarrow \infty.$$

Cramér–Rao lower bound:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

For N independent observations:

$$I_N(\theta) = N I_1(\theta).$$

For linear models with Gaussian white noise:

$$\hat{\theta}_{\text{LS}} = \hat{\theta}_{\text{ML}}.$$

18 18. Identification Workflow

Experiment → Model Structure → Parameter Estimation → Validation → Refinement

Experiment:

- sufficient excitation;
- relevant frequency content;
- adequate sampling;
- avoid saturation/deadzones;
- handle offsets/trends.

Model structures:

$$A(q)y(k) = B(q)u(k) + e(k)$$

(ARX)

$$A(q)y(k) = B(q)u(k) + C(q)e(k)$$

(ARMAX)

19 19. Residual Validation

Residual:

$$r(k) = y(k) - \hat{y}(k).$$

A good model should leave residuals without systematic predictable structure.

Check:

- residual mean:

$$\bar{r} \approx 0$$

- residual autocorrelation:

$$E[r(k)r(k - \tau)] \approx 0$$

for relevant nonzero τ ;

- residual/input correlation:

$$E[r(k)u(k - \tau)] \approx 0;$$

- performance on independent validation data.

zero-mean residuals alone do NOT prove the model is correct.

20 20. Training vs. Validation

Training data:

estimate parameters

Validation data:

test generalization.

A model with excellent training fit but poor validation performance may be overfitting.

Prefer the simplest model that explains the relevant dynamics.

21 21. AIC and BIC

AIC:

$$\text{AIC} = 2p - 2 \log L_{\max}$$

BIC:

$$\text{BIC} = p \log N - 2 \log L_{\max}$$

Lower is preferred.

Both balance goodness of fit against model complexity.

BIC has the stronger complexity penalty as N increases.

22 22. Physical Interpretation

General identification clues:

- slow response $\rightarrow$ dominant pole close to stability boundary;
- fast response $\rightarrow$ faster poles;
- oscillatory response $\rightarrow$ complex-conjugate poles;
- inverse response $\rightarrow$ relevant zero structure;
- unnecessarily complex model $\rightarrow$ possible overfitting.

Choose the simplest physically meaningful model that explains the data.

23 23. MAP Decision Tree

Unknown parameters + data	$\Rightarrow$ LS
Data arriving recursively	$\Rightarrow$ RLS
Forgetting factor	$\Rightarrow$ RLS with forgetting
Can parameters be identified?	$\Rightarrow$ rank/determinant of R_N
Input richness?	$\Rightarrow$ persistent excitation
Is noise white?	$\Rightarrow$ check lagged covariance
LS bias?	$\Rightarrow$ check $E[\varphi v]$
Probability density given?	$\Rightarrow$ ML
Independent Gaussian sensors?	$\Rightarrow$ inverse-variance weighting
Model validation?	$\Rightarrow$ residuals + validation data
Model order?	$\Rightarrow$ validation + AIC/BIC.

24 24. Formula Box – Absolute Essentials

$$\hat{\theta}_{\text{LS}} = (\Phi^\top \Phi)^{-1} \Phi^\top Y$$

$$R_N = \Phi^\top \Phi = \sum \varphi(k) \varphi^\top(k)$$

$$\text{rank}(\Phi) = n_\theta \iff \text{unique identification}$$

$$K(k) = \frac{P(k-1)\varphi(k)}{\lambda + \varphi^\top(k)P(k-1)\varphi(k)}$$

$$\varepsilon(k) = y(k) - \varphi^\top(k)\hat{\theta}(k-1)$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\varepsilon(k)$$

$$P(k) = \frac{1}{\lambda} [I - K(k)\varphi^\top(k)]P(k-1)$$

$$N_{\text{eff}} \approx \frac{1}{1 - \lambda}$$

$$\hat{\theta}_{\text{ML}} = \frac{\sum y_i / \sigma_i^2}{\sum 1 / \sigma_i^2}$$

$$\text{Var}(\hat{\theta} - \theta) = \frac{\sigma^2}{N}$$

$$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{N} \sum y_i^2$$

$$v(k) = e(k) + ce(k-1) \Rightarrow E[v(k)v(k-1)] = c\sigma_e^2$$

$$E[\varphi(k)v(k)] = 0 \Rightarrow \text{no asymptotic LS bias under the relevant assumptions}$$

$$\text{AIC} = 2p - 2 \log L_{\text{max}}$$

$$\text{BIC} = p \log N - 2 \log L_{\text{max}}$$

25 25. Full-Score Checklist

Before submitting a Module 3 answer:

- ☐ Did I write $y = \varphi^\top \theta + e$?
- ☐ Did I get every sign in φ correct?
- ☐ Did I write the LS cost when required?
- ☐ Did I write/derive the normal equations?
- ☐ Did I check rank/determinant if identifiability matters?
- ☐ Did I distinguish $\Phi^\top \Phi$ from $\Phi^\top Y$?
- ☐ For RLS, did I calculate ε first?
- ☐ With forgetting, did I include λ in both K and P ?
- ☐ Did I remember $N_{\text{eff}} \approx 1/(1 - \lambda)$?
- ☐ For whiteness, did I explicitly calculate a lagged covariance?
- ☐ For LS bias, did I check $E[\varphi v]$?
- ☐ For ML, did I construct the likelihood first?
- ☐ Did I use inverse-variance weighting for Gaussian sensors?
- ☐ Did I use N , not $N - 1$, for Gaussian ML variance?
- ☐ Did I distinguish training and validation data?
- ☐ Did I check residual mean, autocorrelation and input correlation?
- ☐ Did I justify conceptual True/False answers?

□ Did I show enough algebra for numerical answers?

26 26. Final Exam Priority

If time is limited, master these in this order:

